ITPFA0-44 PROJECT

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1. Project

Project

Faculty:	Information Technology
Module Code:	ITPFA0-44
Module Name:	Python for Data Science
Content Writer:	Doreen Muderede
Internal Moderation:	Community of Practice
Copy Editor:	Phindile Tabata
Total Marks:	100
Submission Week:	Week 6

This module is presented on NQF level 5.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual assignment.

This assignment 30% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

2. Instructions to Students

Instructions to Students

1. Please ensure that you use the Assignment Template when preparing your submission (where applicable).
2. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
3. Remember to keep a copy of all submitted assignments.
4. All work must be typed.
5. Please note that you will be evaluated on your writing skills in all your assignments.
6. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
7. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

3. Section A

Section A

AI Use Requirements for This Assessment

This assessment is classified as AI-1 as defined in the AI Use Category Table below.

AI Use Category Table

For this specific assessment, the following applies:

Code	Category	What This Means	Evidence Required	Disclosure Requirement
AI-1	Limited AI Use	You may use AI only for simple support tasks (e.g. basic language and grammar, clarifying a definition or generating synonyms). The planning, brainstorming, research and final output must still be your own.	You will be required to attach full transcripts of all your AI chats for evidence of your process.	You must complete and submit an **AI Disclosure Appendix** explaining how you used AI. Access the AI Disclosure Appendix Template here.

4. Terms of AI use for this Assessment

Terms of AI use for this Assessment

1. What Counts as AI?

AI includes standalone or embedded generative systems that create or materially transform assessed work.

This includes AI features embedded in browsers, search engines, office applications, development environments, coding assistants, data platforms, security tools and engineering software.

The product name is less important than what the feature does.

See What Counts as AI

2. Permitted AI Activities

AI may suggest search terms, but it may not locate, select, summarise or evaluate sources for you.

See Permitted AI Activities

3. Assessed Concepts and General Terminology

You may not use AI to explain an assessed module concept as a substitute for engaging with the approved learning material. Each assessment will distinguish between:

Category	Meaning
Assessed concepts	Module knowledge, methods, technical processes or decisions that you must understand and demonstrate independently
General academic terminology	Terms such as peer review, Boolean operator, introduction or primary source that are not themselves being assessed
Borderline terminology	A term whose status is unclear and must be confirmed with the lecturer before using AI

4. Generally Prohibited Activities

These restrictions apply only where the activity forms part of the published assessment criteria. Assessment-specific instructions take precedence where they expressly permit or provide a tool, dataset, configuration, model, code component or other output. Please refer to the below and apply as appropriate.

See Generally Prohibited Activities

4.1. Discipline-specific Prohibited Activities

Please ensure that you consult the prohibited activity list that is specific to this module's discipline from the four discipline options below.

- 4.1.1. Software Engineering
- 4.1.2. Network Security Engineering
- 4.1.3. Data Science
- 4.1.4. Robotics

5. Question 1

Question 1 — 100 Marks

Study the scenario and complete the question(s) that follow(s):

Siyahamba Taxi Association

Siyahamba Taxi Association is a fictional minibus taxi association operating from five ranks in Gauteng: Bree, Noord, Bara, Mabopane, and Germiston. The association's operations manager wants to better understand trip performance across ranks, vehicle types, and time, in order to make informed decisions about fleet allocation, driver scheduling, and peak-hour operations.

You have been provided with six months of trip-level data (July – December 2025) in the file siyahamba_taxi_trips.csv. You will use this dataset to complete the tasks below.

Question 1 (20 Marks)

1.1. Convert the Passengers column into a NumPy array and calculate the total, mean, minimum, and maximum number of passengers per trip. (4 marks)

1.2. Create a NumPy array of the TripRevenue_ZAR column and use array slicing to extract the first 15 values and every fifth value from the last 50 values. (4 marks)

1.3. Use NumPy to calculate the standard deviation and variance of Distance_km. (4 marks)

1.4. Create two NumPy arrays (e.g. Passengers and Fare_ZAR for the first 20 rows) and use element-wise multiplication to manually recalculate trip revenue, then compare your result to the TripRevenue_ZAR column. (4 marks)

1.5. Reshape a NumPy array of your choice (e.g. the first 24 values of Passengers) into a different shape (such as 6 x 4) and explain, in a comment, what reshaping achieves. (4 marks)

Question 2 (25 Marks)

2.1. Load siyahamba_taxi_trips.csv into a pandas DataFrame and display the last 10 rows, the data types of each column, and summary statistics using .describe(). (5 marks)

2.2. Identify and report how many missing values exist in each column. (3 marks)

2.3. Handle the missing values in Rainfall_mm and PassengerRating using an approach of your choice (e.g. filling with the mean/median, or dropping rows), and briefly justify your choice in a comment or markdown cell. (5 marks)

2.4. Filter the DataFrame to show only trips from the Bara rank where PeakHour is equal to 1. (4 marks)

2.5. Group the data by VehicleType and calculate the total TripRevenue_ZAR and average Passengers per vehicle type. (4 marks)

2.6. Add a new calculated column, FuelCost_Est, estimated as 25% of TripRevenue_ZAR, and display the five longest trips by Distance_km. (4 marks)

Question 3 (20 Marks)

3.1. Create a simple line plot showing total weekly trip revenue across the six-month period. (8 marks)

3.2. Add a clear title, x-axis label, and y-axis label to your plot. (4 marks)

3.3. Customise the plot with at least one styling choice (e.g. marker style, line colour, figure size, or gridlines) and explain, in a comment, why you chose it. (5 marks)

3.4. Save your plot to an image file (e.g. weekly_revenue.png) using Matplotlib. (3 marks)

Question 4 (25 Marks)

Using Matplotlib (and/or pandas' built-in plotting), create five different charts. For each chart, include a title, axis labels, and one to two sentences interpreting what the chart shows about Siyahamba Taxi Association's operations. (5 marks per chart: 2 marks for the chart, 1 mark for title and labels, 2 marks for the interpretation)

Question 5 (10 Marks)

Conclude question 4 with a short-written summary (150–250 words) of the key insights Siyahamba Taxi Association's management should take from your visualisations

You must submit the following documents

- A single Jupyter Notebook (.ipynb) or well-commented Python script (.py) containing all code, outputs, and written interpretations for question1–4.
- An exported PDF or HTML version of your notebook/python script showing all executed outputs and charts.
- Any saved chart image files referenced in your submission (e.g. daily_revenue.png).

End of Question 1

6. How AI use Affects Marking

How AI use Affects Marking

Finding	Outcome
No AI used and accurately declared	Your lecturer will mark according to the marking guide.
Permitted AI use accurately declared	Your lecturer will mark according to the marking guide.
AI-generated in identifiable assessed component	Your lecturer will mark authentic work and will not award marks for the affected component that is identified to be AI-generated.
AI-generated in dependent criteria may be affected	Your lecturer will evaluate each criterion in isolation and will not award marks for affected criteria that are AI-generated.
Technically correct work that raises authenticity concerns*	Your lecturer will complete the review before finalising the mark, and will not award marks for affected criteria that are AI-generated.
Serious authenticity concerns exist*	Your lecturer may request a formal discussion with you, and depending on the outcome may escalate a case for formal academic-integrity hearing.

*See General Prohibited Activity list under Terms of AI use for this Assessment